

PRODUCTION-DEPLOYED MULTI-TENANT BIOMETRIC IDENTITY SAAS · ONE HOSTED LOGIN BUTTON

<FIVUCSAS/>

Face & Identity Verification using *Cloud-based SaaS* Models

TEAM · DEPARTMENT OF COMPUTER ENGINEERING

Ahmet Abdullah Gültekin	ahmetabdullahgultekin@gmail.com	github.com/ahmetabdullahgultekin	
Ayşe Gülsüm Eren	ayse.gulsum@marun.edu.tr	aysegulsumeren@gmail.com	github.com/aysegulsum
Ayşenur Arıcı	aysenurarici@marun.edu.tr	aysenurarici@hotmail.com	github.com/Aysenur15

SUPERVISOR Assoc. Prof. Dr. Mustafa Ağaoğlu

CSE4298

ENGINEERING PROJECT
Spring 2025 – 2026

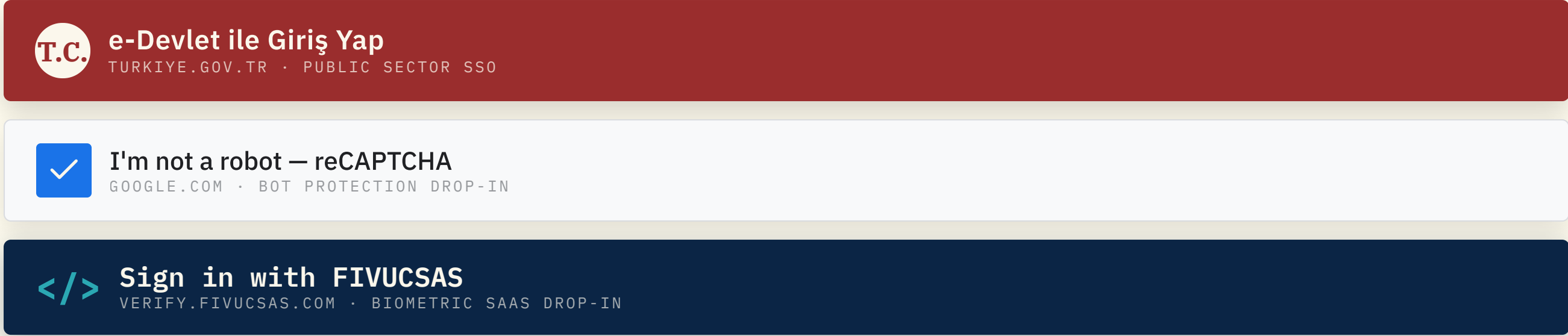
01 Introduction

PROBLEM · WHAT WE BUILT · MAIN GOAL · OBJECTIVES

02 The Drop-in Identity Button

E-DEVLET & RECAPTCHA MENTAL MODEL · SIDE-BY-SIDE · 3 USE-CASES

Three buttons, one mental model

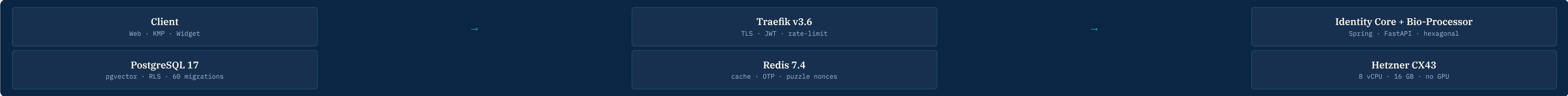


Side-by-side comparison

ASPECT	E-DEVLET	RECAPTCHA	FIVUCSAS
NFC document	—	—	T.C. + ICAO 9303 (mobile) + Web NFC (simple cards)
Integration	Gov approval	Public site-key	Public OIDC + drop-in button
Self-host	—	—	Docker Compose · GPU-less
Open source	—	—	MIT (spoof-detector) · OSS roadmap

03 System Architecture

HEXAGONAL / DDD · 7 CONTAINERS · HETZNER CX43



04 Face Recognition Pipeline — 8 Stages

DETECTION → QUALITY → LANDMARK → ALIGN → EMBED → ENCRYPT → INDEX → MATCH

Server-authoritative pipeline in biometric-processor / (FastAPI, Python 3.12). Every stage is replaceable — backbone, detector and similarity all swap via DI in app/infrastructure/ml/factories/. Verification (1:1) compares against the enrolled vector with cosine distance; identification (1:N) returns pgvector ANN top-k. Aged embeddings (> 2 years) use a more permissive threshold.

01 · DETECT · QUALITY DeepFace 0.0.98 openv · anti-spoof 8=0.5 · Laplacian blur s100 · min 80px · occlusion CIE-Lab quality ≥ 70	02 · LANDMARK · ALIGN MediaPipe Tasks 168-point Facelandmarker · DeepFace eye-line affine 468 + 10 iris	03 · EMBED · ENCRYPT Facenet-512 · Fernet 512-D L2-normalised · AES-128-CBC + HMAC-SHA256 at rest dta = 512	04 · INDEX · MATCH pgvector · cosine HNSW m=16/sf=64 · 1:N ANN top-k · aged-embedding 8=0.38 0 = 0.45
--	---	---	---

05 NFC Document Subsystem — Two Rails

NATIVE MOBILE (ICAO 9303) · WEB NFC (SIMPLE CARDS) · CROSS-CHECK TO LIVE SELFIE

Two rails, one verifier. **Rail A — native mobile NFC**: T.C. ID + ICAO 9303 passports + any BAC-chip document over Android IsoDep + Bouncy Castle. **Rail B — Web NFC**: ISO 14443 simple cards (loyalty, transit, NTAG2xx) directly from Chrome on Android.

T.C. Yeni Kimlik Kartı

The contactless ISO/IEC 14443 chip lives on the back of the T.C. ID card; on passports it is in the cover. Common ICAO AID A0 00 00 02 47 10 01. The same BAC key derivation (SHA-1 from M

reused for TD3 passports and every other ICAO 9303 document carrying the same chip.

Standard: ICAO Doc 9303 (Machine Readable Travel Documents)

Spec ref: practice-and-test/PASSPORT_NFC_ROADMAP.md:8

A. Protocols implemented

- **BAC (Basic Access Control)** — SHA-1 key derivation from MRZ, identical for Turkish T.C. ID and TD3 passports. PASSPORT_NFC_ROADMAP.md:678-690
- **PACE** (Password Authenticated Connection Establishment) — optional for newer passports, scoped not implemented. :164-170
- **Active Authentication · Chip Authentication** — [future work]
- **EF.SOD** validator — 60% complete; CSCA full chain pending. :383-395
- **ICAO check-digit** validation: biometric-processor/tests/unit/test_nfc_mrz_route.py:47-88

B. Data Groups parsed

- **DG1 — MRZ**: .../TurkishEidNfcReader/.../util/Dg1Parser.kt
- **DG2 — Facial image** (JPEG2000 / JP2): Dg2Parser.kt:1-50
- **EF.COM** — data-group presence list
- DG3 / DG4 / DG11 / DG13 / DG14 / DG15: *not implemented (roadmap §5.1)*

06 spoof-detector — Hybrid PAD, Session Verdict, In-Browser

SUBMODULE · MIT · PAPER DRAFT · IBETA L1 SUBMISSION · AMISPOOF.FIVUCSAS.COM

Measured results (paper, bootstrap 95 % CI on 100 stratified resamples)

DATASET / REGIME	PIPELINE	VALUE	95 % CI	SOURCE
CASIA-FASD AUC (zero-shot, N = 2 408)	minifasnet_only	0.9452	[0.9366, 0.9560]	07_results.md:15
Per-frame latency (Hetzner CX43 CPU)	hybrid	63.0 ms	p99 = 117.8 ms	:110
Browser (desktop Chrome, WebGPU)	browser bundle	25 – 30 fps	—	BROWSER_READINESS.md:4
ISO/IEC 30107-3 (in-house scripted)	session	Ggrade C	BPCR 0 % · APCR 30 % · ACER 15 %	README.md:74-84

07 The Biometric Puzzle — 23 Micro-Challenges

14 FACE + 9 HAND-GESTURE · SERVER-NONCED · REPLAY-PROOF · CANONICAL ENUM BIOMETRICPUZZLEID

23 canonical micro-challenges (14 face + 9 hand). Server emits a random UUID-nonce sequence of 2-7 actions per attempt; client streams only landmark vectors. Pre-recorded, AR-filter, deepfake, replay all fail — sequence unpredictable, server-immutable.

14 face + 9 hand challenges

Blink · smile · open mouth · turn L/R · look up/down · raise brows · close eye · nod · shake · finger count · wave · flip · tap · pinch · peek-a-boo · math · shape-trace · template-trace · hold-pose

08 Ten Composable Authentication Factors

NIST 800-63B · KNOW / HAVE / ARE / SHOW · TENANT JSON, NOT JAVA

10 Per-tenant MFA composition · no backend code required									
Password BCrypt-12	Email OTP 6-digit · 5 min	SMS OTP Twilio · 6-digit	TOTP RFC 6238 · 30 s	QR Code cross-device	Face Facenet-512 + Puzzle	Voice Resemblyzer · 256-D	Fingerprint WebAuthn platform	Hardware Key F2002 · WebAuthn	NFC Document ICAO 9303 · BAC

10 Subdomain Ecosystem

9 SUBDOMAINS · 1 QR · ALL BEHIND TRAEFIK V3.6

One QR · all links. A single QR code on this poster points to a **link navigator page we own**: links.fivucasas.com. We do not depend on a third-party shortener — our own static subdomain hosts every project subdomain, the team (GitHub, LinkedIn, portfolio, email), poster artefacts (HTML, PDF, PNG, text brief) and shortcuts to live surfaces. Hostinger static (no Docker), the same pattern as bys-demo / amispoof / landing.



One QR · all links

Scan to open the FIVUCSAS hub

A single QR for landing + hosted-login + admin + docs + demo + spoof-detector + status — every surface under one short link. Scan from any phone and jump to the surface you need.

<https://fivucasas.com/links>

REPOSITORIES

github.com/Rollingcat-Software/fivucasas (private)

github.com/Rollingcat-Software/spoof-detector (MIT)

6 git submodules · 1 self-hosted sumner

LIVE SURFACE

fivucasas.com · verify.fivucasas.com · demo.fivucasas.com

api.fivucasas.com · app.fivucasas.com · docs.fivucasas.com

amispoof.fivucasas.com · status.fivucasas.com

CONTACT

ahmetabdullahgultekin@gmail.com

ayse.gulsum@marun.edu.tr · aysegulsumeren@gmail.com

aysenurarici@marun.edu.tr · aysenurarici@hotmail.com

Supervisor: · ağaoğlu @ marmara.edu.tr

FIVUCSAS · Marmara Üniversitesi · CSE4297 / CSE4298 Engineering Project · Spring 2025 – 2026 · AR poster v3 · evidence-pinned 2026-05-19 · every metric cited to file:line — see POSTER_BRIEF.md at repo root for full source map.